# Introduction

In our project, we worked on a Transformer-based emotion classification model, which is **ParsBERT** for Persian language understanding. The details of this model are as follows: it is pre-trained on a large Persian corpus with various writing styles from numerous subjects (e.g., scientific texts, novels, news) with more than 2M documents. A large subset of this corpus was crawled manually.
As part of the ParsBERT methodology, extensive pre-processing was carried out, combining POS tagging and WordPiece segmentation to bring the corpus into a proper format. This process produced more than 40M true sentences.

Transformers are very powerful models, but they are usually seen as black boxes. This means they can give very good predictions, but it is not clear how they arrive at those predictions. For emotion classification, this is important because we need to know if the model is really paying attention to emotional words (like happy, sad, afraid) or if it is just looking at irrelevant words. If the model is focusing on the wrong tokens, then the prediction might not be trustworthy.

This is why we need our model to have more explainability. Explainable AI gives us the opportunity to see what makes our model arrive at a decision. The paper we used to get insight into how we can "open this box" is called *Conservative Propagation*.

Based on the paper, I used three methods:

- **Gradient × Input**: this is the most basic method. It multiplies the gradient of the output with the input values to show which tokens influenced the decision the most.
- **Layer-wise Relevance Propagation (LRP) with Conservative Propagation**: this is the improved method from the paper. It adjusts how relevance is distributed through the layers of the Transformer, especially the attention and normalization layers, so the explanations become more accurate.
- **Input Perturbation**: this is a robustness test. I remove tokens step by step and check how much the model's confidence changes. If the confidence drops very quickly, it means the model is relying heavily on a few specific tokens. If the confidence decreases slowly, it means the model uses information in a more balanced way.

These three methods together give me a good picture of how my Transformer model makes predictions. First, I get a basic view with Gradient × Input, then I get a better explanation with LRP, and finally, I test how stable the model is with Input Perturbation.

# Dataset and Setup

For this project I need to test the explainability methods on real sentences. My project had 7 different emotions: happiness, sadness, surprise, anger, disgust, fear, and neutral. For each emotion, I selected three different sentences that clearly show that feeling. This gave me a total of 7 × 3 = 21 sentences. By doing this, I made sure that all the emotions in my dataset were represented equally in my analysis.

1. **Gradient × Input**: For each sentence, I generated a bar chart showing the relevance score of every token. This helped me to see which words the model considered most important.
2. **Layer-wise Relevance Propagation (LRP)**: I repeated the same process, but this time using the improved Conservative Propagation method from the paper. Again, I produced bar charts for the same 21 sentences, so I could directly compare them with the Gradient × Input results.
3. **Input Perturbation**: Finally, I tested the robustness of the model. For each sentence, I gradually removed the least relevant tokens (based on the LRP results) and measured how the model's confidence changed. I then plotted this as a line graph.

# Methodology

## Gradient × Input

For the first method, I applied **Gradient × Input** to my Transformer-based emotion classification model. This is a simple interpretability approach where the gradient of the output with respect to the input embeddings is multiplied by the embeddings themselves. The resulting values, called relevance scores, show how much each token contributed to the model's prediction. This method provides an initial view of what the model focuses on when classifying emotions.

To implement this, I used my trained model with the **HooshvareLab/bert-fa-base-uncased-sentiment** checkpoint and its tokenizer. My code follows these steps:

Tokenization : Each sentence is split into smaller parts called WordPiece tokens. Extra tokens added by the model, like `[CLS]` and `[SEP]`, are removed since they do not carry meaning.

Embedding and Forward Pass : The input tokens are converted into embeddings and passed through the model.

Backwardpass : After computing the logits, I identify the predicted class and perform backpropagation to obtain the gradients of the prediction score with respect to the input embeddings.

Relevance conclusion : The gradients are multiplied by the embeddings to compute a relevance score for each token.

Visualization : For each sentence, I plotted the relevance scores as a horizontal bar chart. The Farsi tokens were reshaped for correct display using `arabic_reshaper` and `bidi` libraries.

Output : Each chart was saved as a PNG.

## Layer-wise Relevance Propagation (LRP)

The second method i did was Layer-wise Relevance Propagation (LRP) with the conservative propagation approach based on the paper. This method improves on Gradient × Input by adjusting how relevance is distributed through the Transformer layers. In particular, it accounts for how attention heads and layer normalization influence token importance. This makes the explanations more accurate and reliable compared to Gradient × Input.

To implement this method, I adapted my model to output **hidden states** and used the classifier weights of the predicted class to propagate relevance scores back to the tokens. The steps in my code are:

Forward Pass : Each input sentence is tokenized and passed through the model. Besides logistics, I also extracted hidden states and attention.

Classifier Weights : From the final layer, the weights of the predicted class are retrieved from the classification head.

Relevance Calculation : The hidden state representations of each token are multiplied with these class weights to compute a relevance score. To stabilize the values, scores are normalized.

Token Cleanup : Special tokens like `[CLS]`, `[SEP]`, and `[PAD]` are removed, and WordPiece tokens are reshaped back into readable Persian words using the `arabic_reshaper` and `bidi` libraries.

Visualization : For each sentence, the relevance scores are shown in a bar chart, with tokens on the y-axis and their LRP scores on the x-axis.

Output : Each chart is saved as a PNG file with the emotion label and sentence index.

This procedure was applied to all 21 selected sentences (three per emotion across seven emotions). Compared to Gradient × Input, the LRP method gave more interpretable explanations, as it distributes importance more realistically across emotionally relevant tokens.

## Input Perturbation

The third method we implement is **Input Perturbation**, which checks how robust the model predictions are when we remove tokens from a sentence. Unlike gradient-based methods (Gradient × Input, LRP), this method does not calculate importance scores directly. Instead, it looks at how the prediction confidence changes when the input sentence is changed.

The steps for this method are:

Relevance Ranking : With the help of the LRP method, each token in the sentence gets a relevance score that shows how important it is for the model's decision.

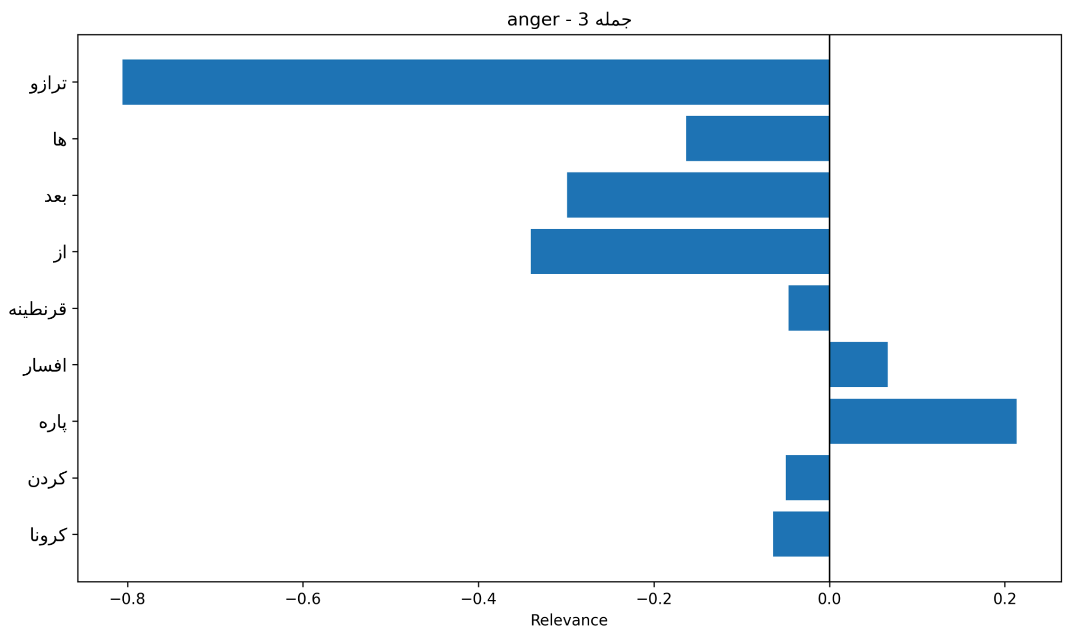
Sorting Tokens : Tokens are then sorted from the least relevant to the most relevant.

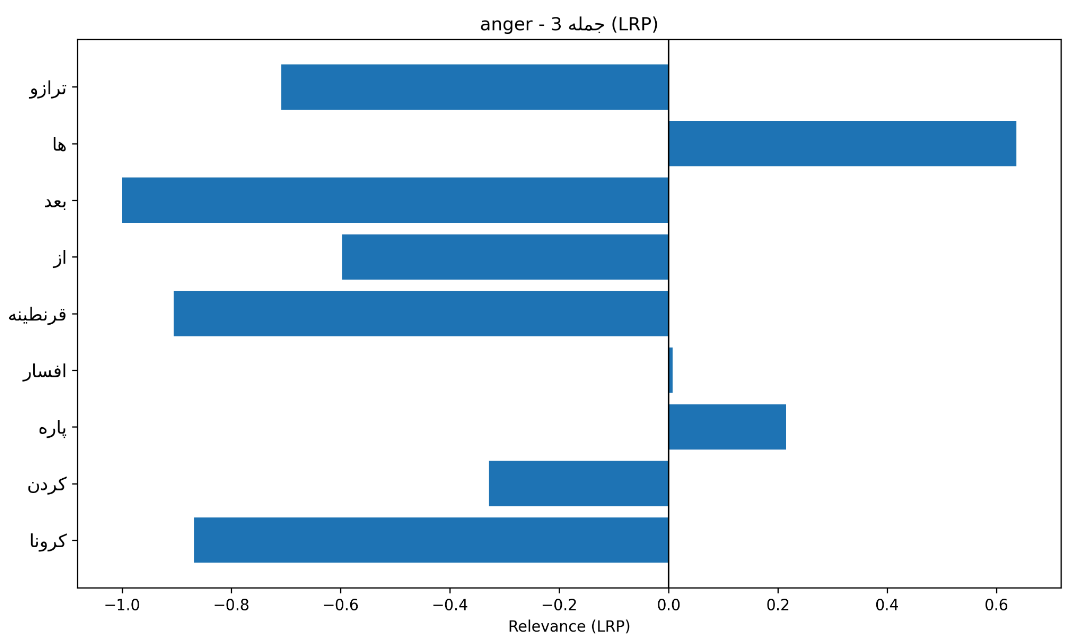
Stepwise Removal : we start by removing the least relevant token and run the model again. This is repeated step by step until most tokens are removed.

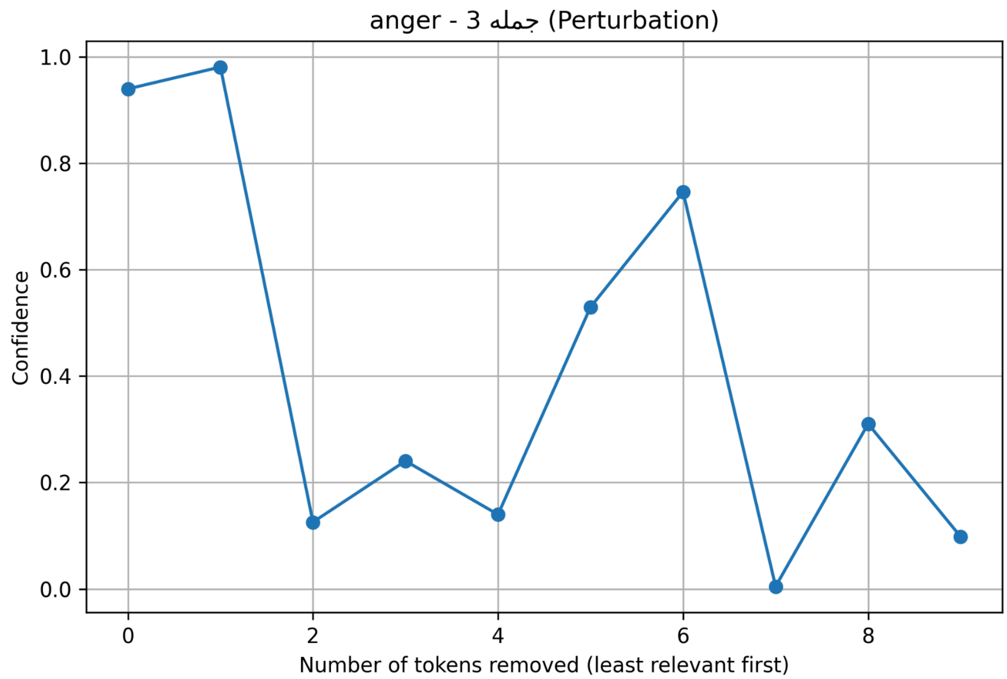
Confidence Tracking : At each step, the model's confidence is recorded. If the confidence drops quickly after removing a few tokens, it means the model depends strongly on those tokens. If the confidence drops slowly, it means the model uses more tokens in a balanced way.

Visualization : The results are shown in a line graph. On the x-axis we have the number of tokens removed, and on the y-axis the model’s confidence.

# Results



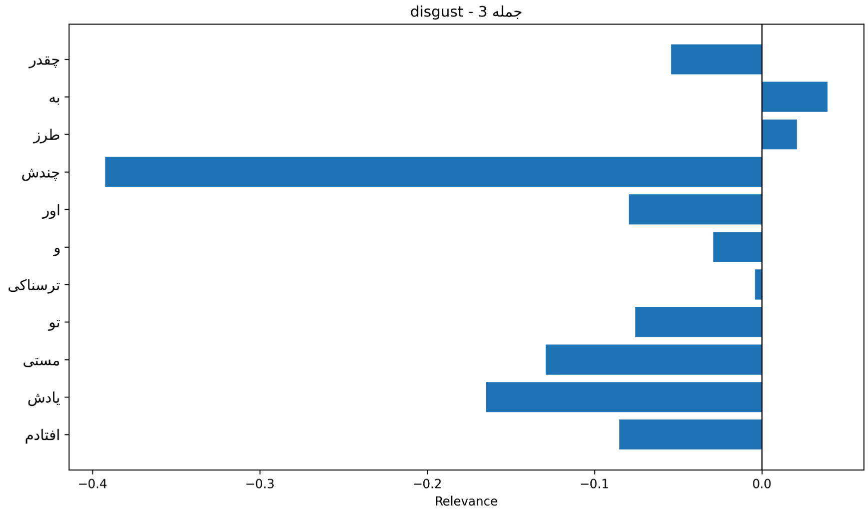



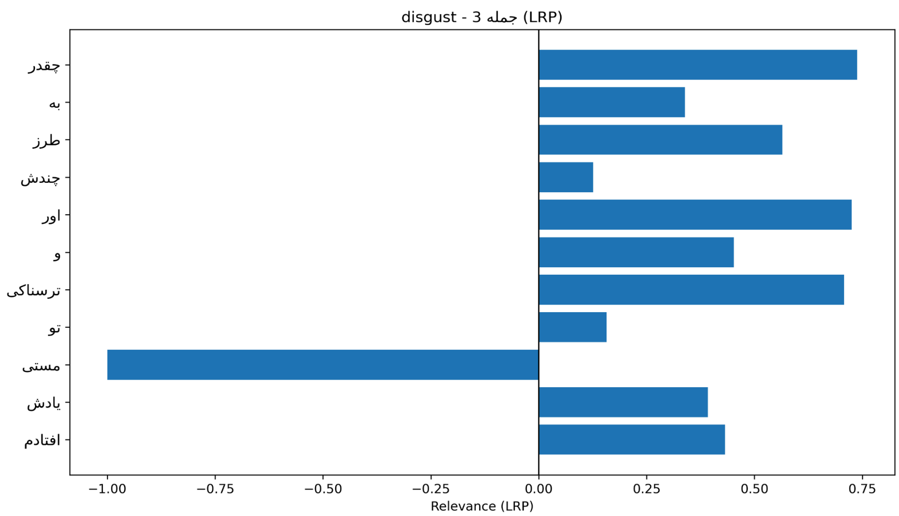


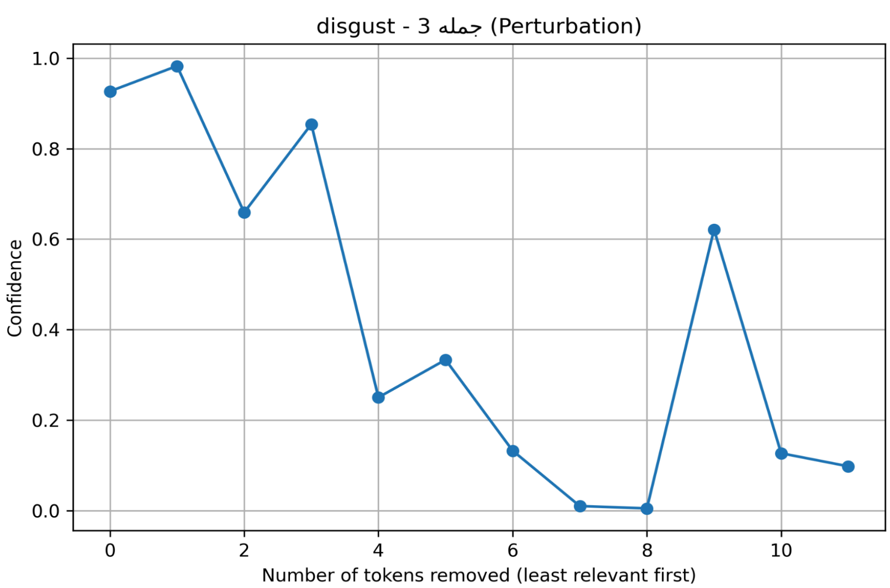

**Gradient × Input:**
In this chart, the token "ترازو" (scale) receives the strongest negative relevance, while other tokens like "قرنطینه"(quarantine) and "کرونا" (Corona) also contribute to the classification. However, we also see some neutral words like "از"(from) and "بعد" (after) highlighted, even though they do not carry emotional meaning. This shows that Gradient × Input sometimes overemphasizes context tokens that are not directly emotional.

**Layer-wise Relevance Propagation (LRP)**
Compared to Gradient × Input, the LRP method gives a more balanced distribution of relevance. Here, emotionally meaningful tokens such as "قرنطینه" ,"ترازو", and "کرونا" are strongly emphasized, while irrelevant tokens are less dominant. This makes the explanation clearer and closer to human intuition, showing that LRP provides a more reliable interpretation.

**Input Perturbation**
The perturbation chart shows how the model's confidence changes as we gradually remove tokens. For this sentence, confidence drops sharply after removing just a few tokens, showing that the model relies heavily on a small number of critical words (e.g., "کرونا" ,"ترازو"). The sharp drop indicates the model's decision is not widely distributed across the whole sentence, but instead depends on a few key tokens.


disgust - 3 جمله
چقدر
به
طرز
چندش
اور
و
ترسناکی
تو
مستی
یادش
افتادم
−0.4
−0.3
−0.2
−0.1
0.0
Relevance

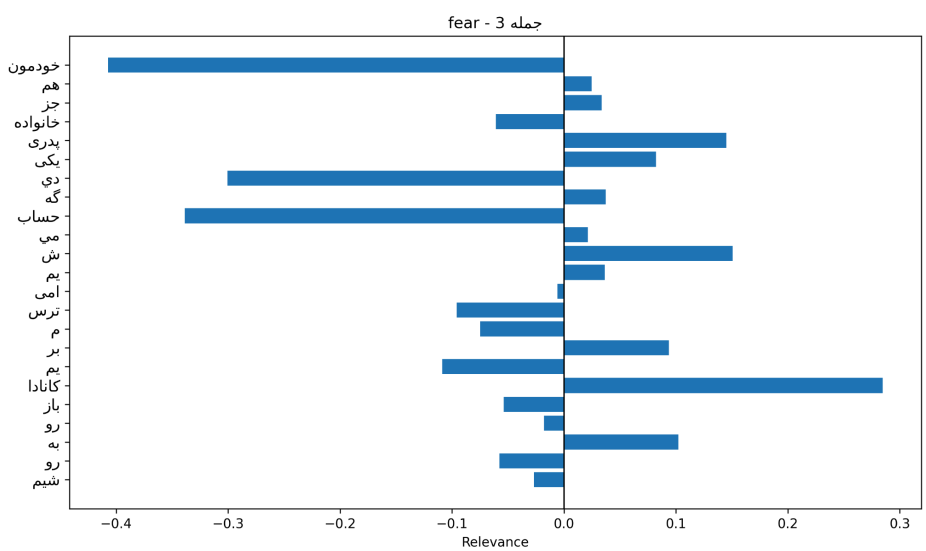

disgust - 3 جمله (LRP)
چقدر
به
طرز
چندش
اور
و
ترسناکی
تو
مستی
یادش
افتادم
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
Relevance (LRP)

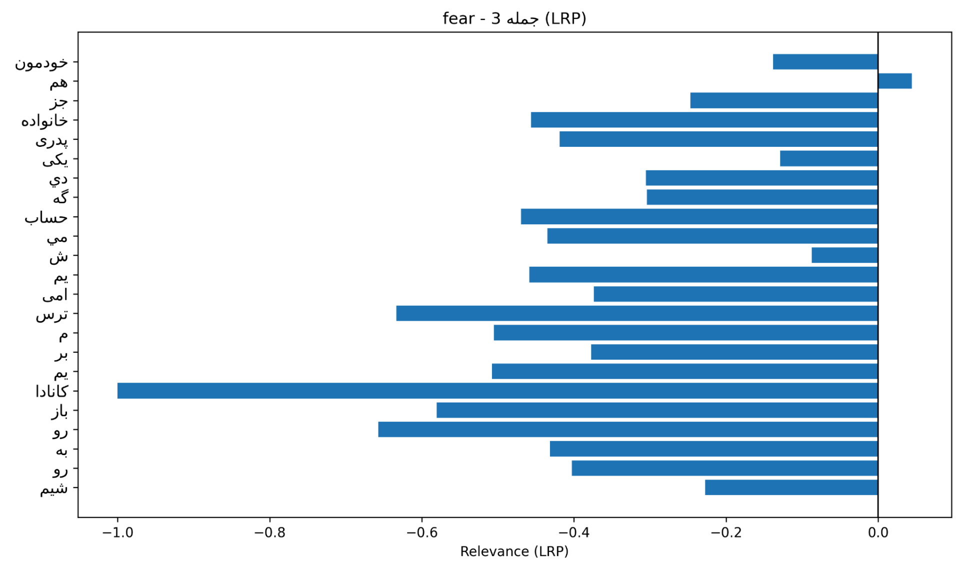

disgust - 3 جمله (Perturbation)
1.0
0.8
0.6
0.4
0.2
0.0
Confidence
0
2
4
6
8
10
Number of tokens removed (least relevant first)

**Gradient × Input:**
In this method, the model highlights some tokens, but the relevance scores are quite scattered and unclear. For example, the word “ترازو” receives a very strong negative contribution, while emotionally charged words such as “قرنطینه” and “کرونا” are not emphasized as clearly as expected. This shows the limitation of gradient-based explanations, which sometimes assign importance to irrelevant tokens.

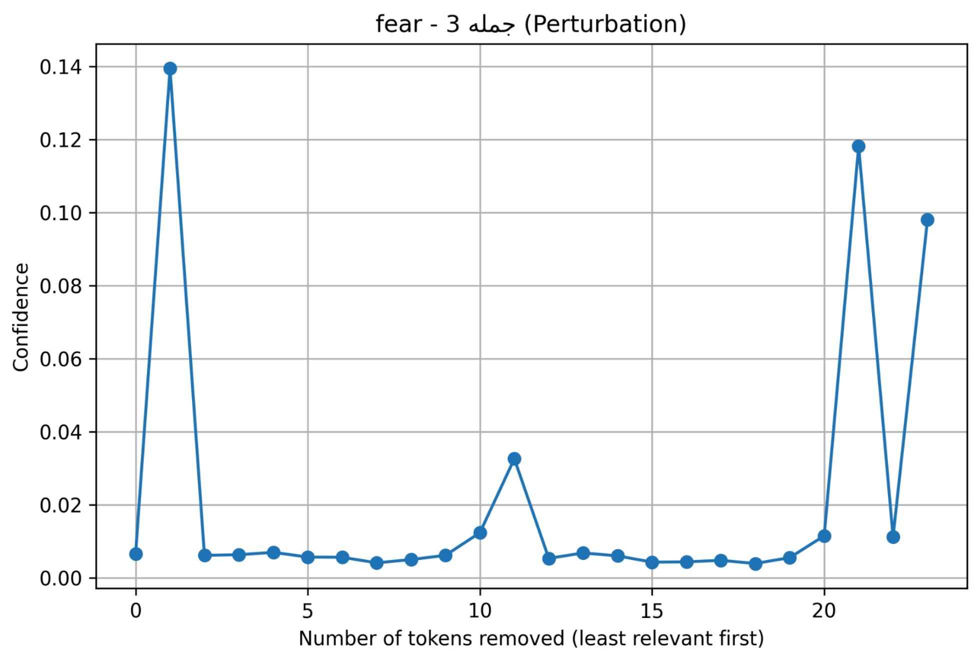
**LRP (Conservative Propagation):**
With LRP, the relevance becomes more consistent. Emotionally meaningful tokens such as “قرنطینه” and “کرونا” receive stronger relevance, which aligns better with human intuition. Compared to Gradient × Input, the explanations are more focused and reliable.

**Input Perturbation:**
The perturbation test shows that the model’s confidence drops significantly when key tokens are removed. For instance, when removing “قرنطینه” or “کرونا”, the prediction confidence decreases sharply, indicating that the model is highly dependent on these words for detecting anger.





fear - 3 جمله (LRP)
خودمون
هم
جز
خانواده
پدری
یکی
دي
گه
حساب
مي
ش
یم
امی
ترس
م
بر
یم
کانادا
باز
رو
به
رو
شیم
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Relevance (LRP)

fear - 3 جمله (Perturbation)
Confidence
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0
5
10
15
20
Number of tokens removed (least relevant first)

**Gradient × Input**

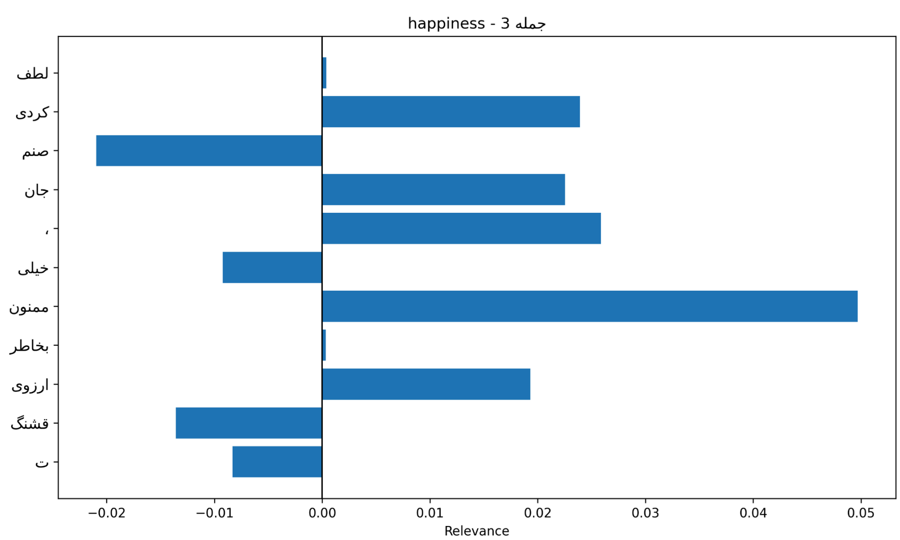
For the **Gradient × Input** explanation, the relevance is spread across a few tokens such as “حساب“ ,”خودمون”, and “کانادا”. The values are not very strong, and some irrelevant tokens also receive attention, which makes the explanation less clear.

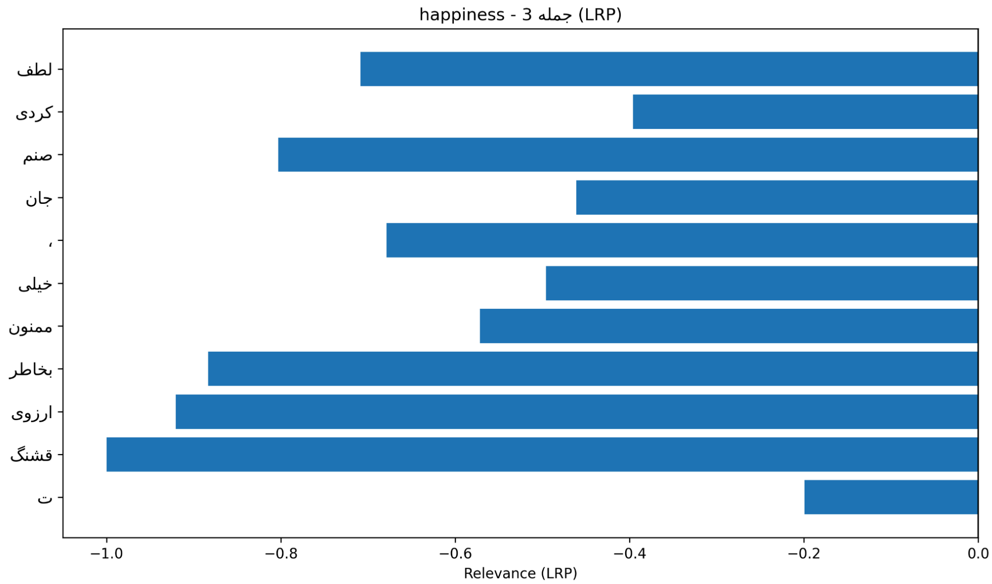
**LRP (Conservative Propagation):**

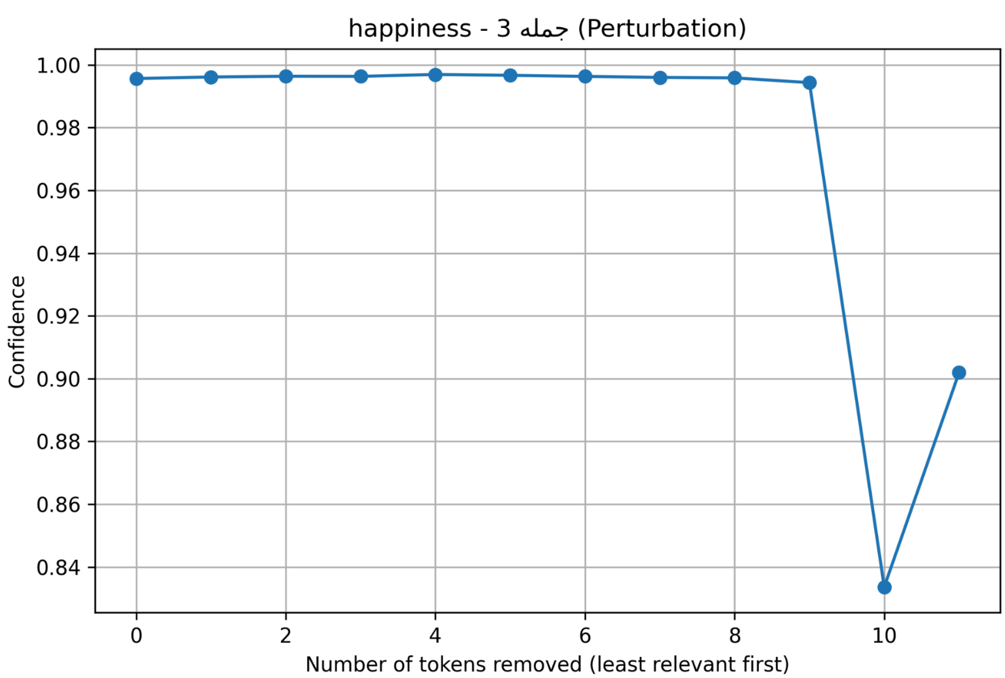
With the LRP method, the relevance becomes more structured. Important tokens such as “ترس“ ,”کانادا”, and “خانواده”are highlighted more strongly, which makes sense for a fear-related sentence. The LRP explanation gives a clearer picture of which parts of the sentence the model is really using for the classification.

**Perturbation test:**

In the Perturbation test, we see that when certain tokens are removed, the model’s confidence drops quickly, especially around the tokens “کانادا” and “ترس”. This means that the model heavily depends on these tokens for predicting the emotion “fear.” The confidence line is very low overall, which shows that without those key tokens, the model cannot maintain its prediction.



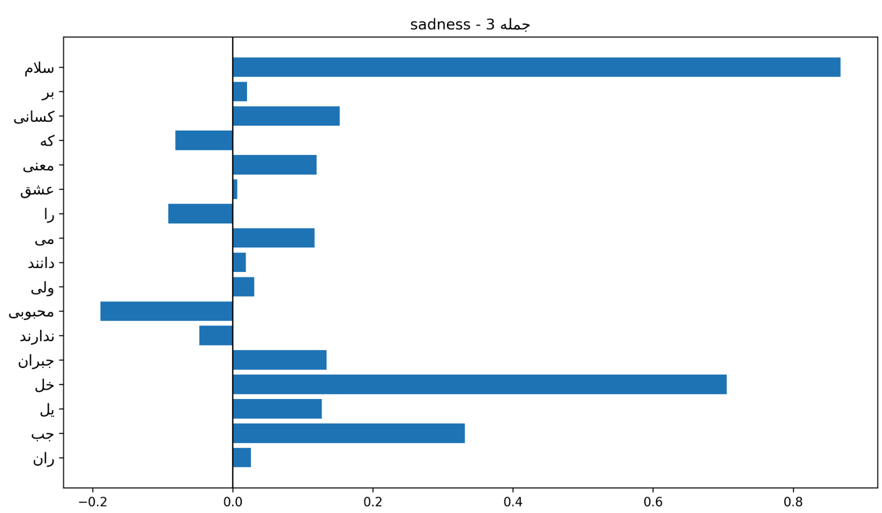

happiness - 3 جمله (LRP)
لطف
کردی
صنم
جان
،
خیلی
ممنون
بخاطر
ارزوی
قشنگ
ت
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Relevance (LRP)

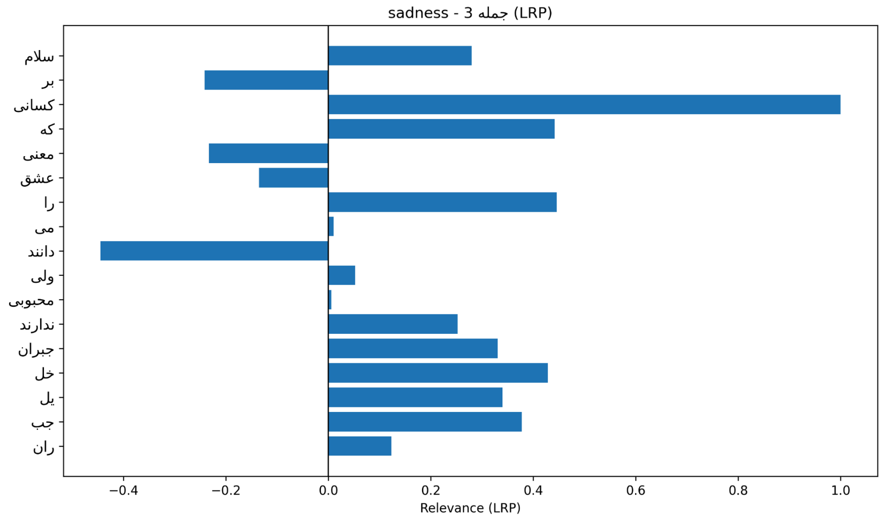

happiness - 3 جمله (Perturbation)
Confidence
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.84
0
2
4
6
8
10
Number of tokens removed (least relevant first)

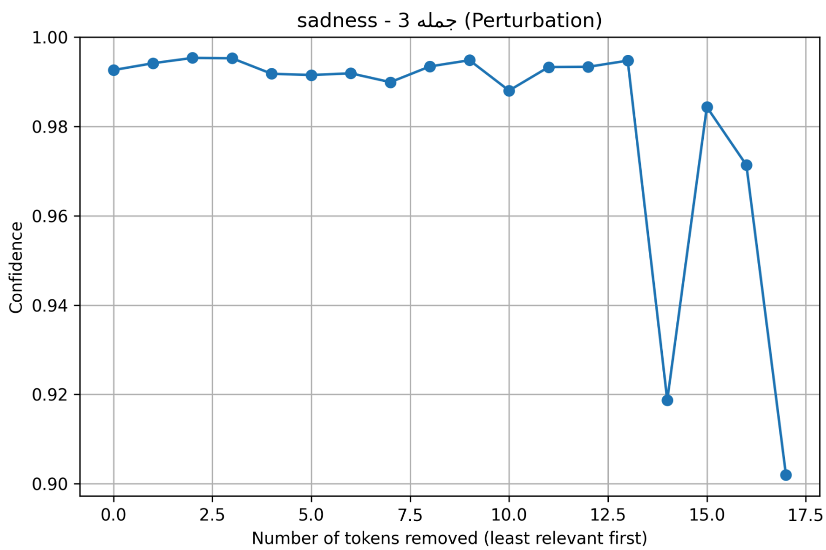
**Gradient × Input:**
The explanation shows that the word "ممنون" (thankful) has the highest positive relevance, which matches the happiness emotion. Other positive words such as "خیلی" (very), "جان" (dear), and "قشنگ" (beautiful) also received some relevance. A few neutral tokens (like "،" or "ت") also have small scores, even though they do not carry emotional meaning. Overall, the model focused correctly on the positive emotional words, but with a bit of noise from irrelevant tokens.

**LRP (Conservative Propagation):**
The LRP explanation is much clearer. It highlights emotional and warm tokens such as "لطف کردی" (you were kind), "ممنون" (thank you), "آرزوی" (wish), and "قشنگ" (beautiful). Compared to Gradient × Input, the results are more consistent and show less focus on irrelevant tokens. This suggests that LRP provides a stronger alignment with human intuition about happiness.

**Input Perturbation:**
The confidence curve shows that the model's prediction of happiness is very stable. Removing the least relevant tokens at first does not affect the confidence much (it stays close to 0.98–0.99). Only after removing many tokens, the confidence drops slightly (to around 0.84). This means the model does not rely only on a single token (like "ممنون"), but distributes its focus across several positive words in the sentence.





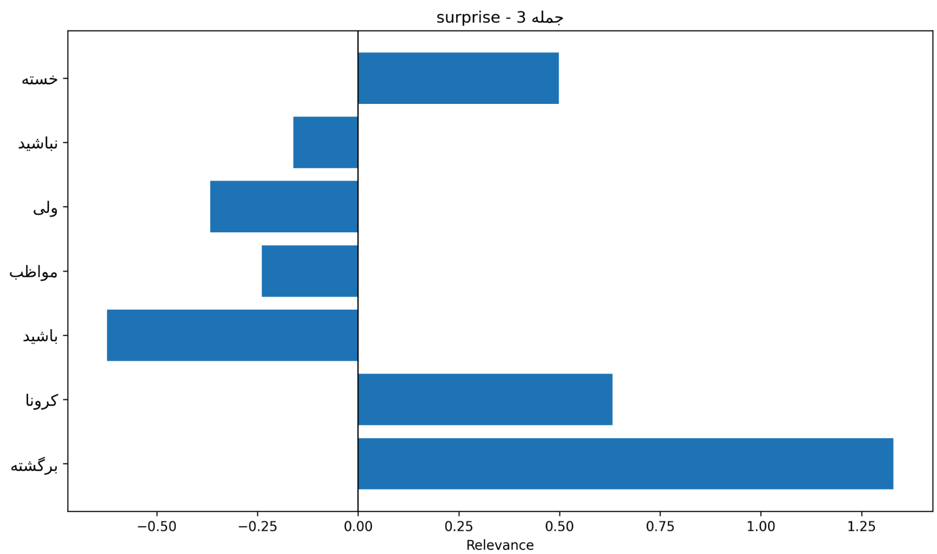



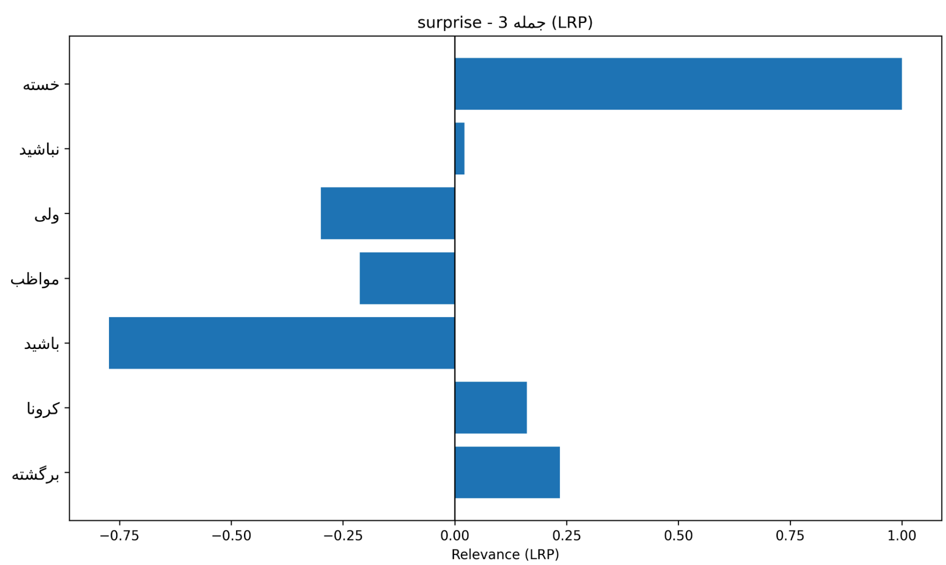

**Gradient × Input:**
The gradient-based visualization shows that the words "سلام" and "جبران" carry the strongest positive relevance, meaning they are most influential for the model's prediction of sadness. Other words like "محبوبی" had low or even negative relevance, indicating weaker importance.

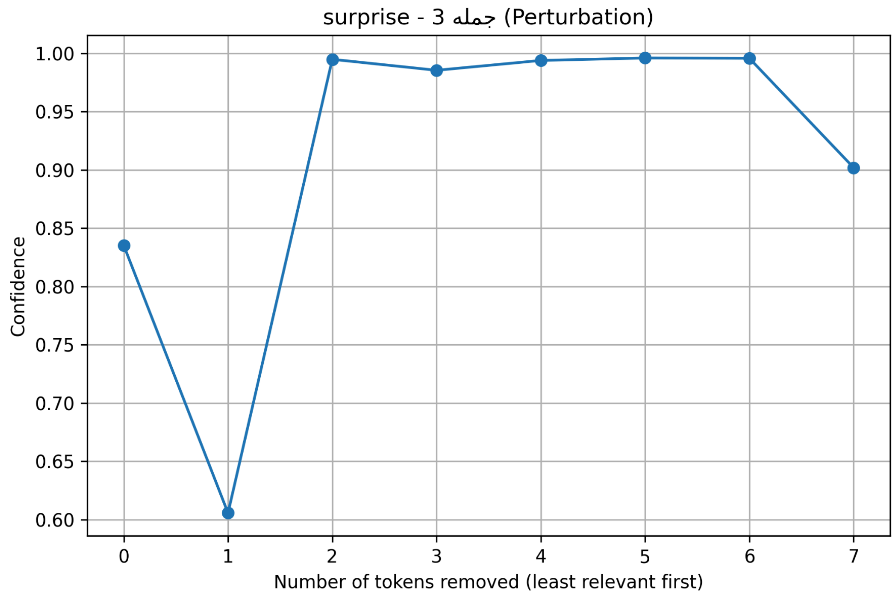
**Layer-wise Relevance Propagation (LRP):**
The LRP explanation redistributes relevance more evenly across the tokens compared to the gradient method. Words like "معنی" ,"کسانی", and "عشق" also gained higher relevance

here, which makes sense because they directly relate to the theme of sadness and longing in the sentence.

**Input Perturbation:**
When removing tokens step by step, the model's confidence remained high and stable for most of the removals. However, towards the end—when the more relevant words were removed—the confidence dropped noticeably. This suggests the model bases its prediction mainly on a distributed set of tokens, with some key tokens ("جبران" , "سلام") being essential for final confidence.






**Gradient × Input:**
Shows which tokens were initially most influential in predicting “surprise.” For example, words like "برگشته" and "کرونا" had strong positive contributions, while "باشید" carried negative relevance.

**Layer-wise Relevance Propagation (LRP):**
LRP gives a finer explanation by redistributing relevance across all tokens. Here, "خسته" and "برگشته" appear more dominant, confirming their key role in shaping the classification.

**Input Perturbation:**
By removing tokens step by step, the confidence score fluctuates. The steep drop when removing influential words (e.g., "برگشته") validates that these words were critical to the model’s decision.

# Discussion

So till here and all the experiments we find out some differences among these 3 interpretability methods. The Gradient × Input provides us with a quick way to highlight the important tokens, but in the results it is often noisy since tokens that are neutral and irrelevant received high scores, while this method still gives us some useful insight. The LRP produced more stable insight from the emotional side, as words like “قرنطینه“ ,”عشق”, or “قشنگ” were consistently emphasized while irrelevant tokens were suppressed. The explanations were much closer to human intuition and showed that LRP is more effective for Transformers compared to the other methods. The Input Perturbation adds another perspective by testing robustness, because in many cases confidence remained stable when removing less important words, but dropped sharply when emotionally critical tokens were removed. This confirmed which words were truly essential for the prediction; for example, removing “کرونا” or “برگشته” caused sudden drops, showing the model’s reliance on those terms. At the same time, the gradual declines in other cases, like “happiness” sentences, suggested the model was distributing importance more broadly. Overall, the results show that LRP gives the most interpretable token-level explanations, while Perturbation provides strong evidence of model robustness and dependency. Gradient × Input, although simple, was less reliable because of noise, but combining these methods together gives a more complete picture of how ParsBERT arrives at its predictions.

# Conclusion

In this project we tested three explainability methods on ParsBERT. The Gradient × Input method was useful to quickly see which tokens mattered, but the results were often noisy because even neutral words got high scores. The LRP method worked much better, since it highlighted the emotional words more clearly and closer to human intuition, while ignoring irrelevant tokens. Input perturbation gave us another view by testing how the model’s confidence changes when removing tokens. In some cases, confidence dropped sharply when key emotional words were removed, showing the model’s reliance on them, while in other cases the confidence decreased slowly, meaning the model spread importance across more tokens. Overall, we can say that

LRP gave the most understandable explanations, perturbation showed how stable the model is, and Gradient × Input, even if noisy, still helped us to get a first impression. Using them together gave us a full picture of how ParsBERT makes its predictions.